Qn			Answers	Marks
6	(a)		It is the probability per unit time of a decay.	B1
	(b)	(i)	half-time = 14 mins = 840 s	B1
			activity = $\frac{\ln 2}{T_{1/2}} N = \frac{\ln 2}{840} \times 2.0 \times 10^7$	B1
			= 16500 Bq or 17000 Bq (2 sf)	A1
			alternative	
			activity = gradient of *N* - *t* graph	B1
			method to find gradient - Correct coordinates - Correct formula	M1
			A between $\pm 10\%$ of 16500 Bq (14900 Bq to 18200 Bq)	A1
		(ii)	half-life is short, so no time for technetium-101 to be delivered into tumour or half-life in short implies activity is high during delivery to tumour. So another organs may be damaged during delivery / too low at the end and therefore ineffective	B1

			or half-life is short, so no time to be technetium-101 to be delivered to hospital after production or half-life is short, power changes quickly over time. Cannot estimate the dose.	